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Does Market Microstructure affect Time-varying Efficiency? Evidence from Emerging Markets

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Abstract

Market microstructure is concerned with the process by which investors' latent demands are translated into prices and volumes. This study tests the effect of seven key market microstructure components: liquidity, volatility, trading cost, thin trading, anonymous trading, automation, and tick size, on time-varying informational efficiency. The dataset comprises of firm level data from five emerging markets, for 2001-2020. Using a system GMM model while controlling for, we find that microstructure, specifically, liquidity, volatility, thin trading, and trading cost are the key determinants of time varying efficiency in emerging markets. In comparison anonymous trading, tick size, and automation are not found to have a significant influence. We also observed heterogeneity at country level results, finding that SSE and NSEI are more efficient. Individual and institutional investors should consider market microstructure components while designing asset allocation strategies.

Key words: Market Microstructure, Market Efficiency, Emerging Asian Markets.

Introduction

According to Fama (1970), a market is efficient if current prices reflect all the available information in a timely manner, which makes it difficult to beat the market on a regular basis. In practice, information efficiency is time-varying, and equity prices react differently to the nature of information. This raises an important question: what are the underlying determinants that affect the informational efficiency of equity prices? Researchers tried both bottom-up (financial statements) and top-down (macro and micro-level indicators) approaches to explain variation in prices. These approaches have limitations in reflecting investors' subjective assessments of market reactions to newly arrived information. Understanding market microstructure has enabled investors to gain valuable insight into the relationship between price and resource allocation. Microstructure facilitates the price discovery process through regulation and systematic trading mechanisms (Bekaert and Harvey, 2003). A well-structured market facilitates information flow, order placement, and execution. This implies that the market microstructure needs to be optimal so that it can address complex market dynamics and changing market regimes. Market reforms, investor-friendly trading mechanisms, and lower transaction costs improve market liquidity (Chordia et al. 2008; Chung and Hrazdil, 2010; and Arjoon, 2016). It also determines the ability of market participants to incorporate information flows into trading orders, thereby increasing the efficiency of the market. The baseline model in this study tests the effect of liquidity (AMIVEST liquidity ratio and stock turnover) and volatility (EGARCH and TGARCH) on informational efficiency.

Prior studies have investigated the effect of different microstructure components on the level of informational efficiency; see, for example, Davila and Parlato (2021) on trading cost, Harrison and Moore (2012) on thin trading, Chung and Hrazdil (2010) on liquidity, and Amihud et al. (2006) on volatility. However, most previous studies focused on a single exogenous variable, while the results of models can be partially influenced by other factors. Therefore, we made significant changes to our baseline model by including portfolio-specific and country-level controlling variables. Furthermore, only a few studies highlight this problem in emerging markets (e.g., Bekaert and Harvey, 2003; Arjoon, 2016; and Arjoon et al. 2020). Unlike developed markets, emerging markets are in the embryonic stage and have undergone a series of microstructure reforms in the last few decades, thus making them an ideal avenue for investigation (Bekaert and Harvey, 2003; Arjoon et al. 2020). The empirical findings of this study are based on five Asian emerging markets: China (SSE), Pakistan (PSX), India (NSEI), Turkey (BSI), and Malaysia (KLSE).

Unlike Arjoon (2016), who considers the bid-ask spread as a proxy for liquidity, this study focuses on Amivest liquidity to analyze the speed at which information is incorporated into stock prices. It also helps to evaluate the dollar value of trading as a result of price changes. We know of the importance of stock turnover, and hence we consider its impact in the robustness analysis. Investor expectations and trading decisions cause prices to move from one equilibrium to another. If the information is timely incorporated into equity prices, it will not cause volatility spikes. Volatility is an indicator of stock market activity and drives time variation in informational efficiency. High volatile markets deter informational efficiency. Price is a dynamic concept and any model that relies only on liquidity results in lower explanatory power (Bekaert and Harvey, 2003). Following Arjoon (2016) and Raza et al. (2022), we also consider volatility in our model.

Markets have traditionally been used as a centralized open-outcry system intended to enable trading among investors. Stock markets have introduced electronic trading systems to facilitate high trading volumes and improve efficiency (Huang et al. 2019). Automation reduces time lag and transaction costs and facilitates improved price discovery (Boehmer and Kelley, 2009). In the last couple of decades, all the emerging markets in our sample have minimized floor trading by automating the market. Electronic trading systems have significantly reduced transaction costs, which are a direct constraint on trading and arbitrage activity. Given that financial markets play an essential role in generating and aggregating dispersed information, this study also tests the effect of automation and transaction costs on informational efficiency.

As shown, a lower tick size reduces pricing error and promotes market quality via an improved price discovery process (Martinez and Tse, 2019). The primary motivation for including tick size in the model is our expectation regarding the positive effect of lower tick size on market quality. The effect of tick size on market efficiency can also be influenced by rules and regulations, trading technology, and market structure. To fully comprehend these dynamics, more investigation and empirical analysis are required. Smaller tick sizes encourage information-based trading; post-decimal prices are likely to be more informative than pre-decimal prices. The absence of a price change between two moments can also be interpreted as being caused by the absence of a price reaction to new information and consequently as a sign of inefficiency, whereas it may simply be driven by the absence of any frequent trade. Thin or infrequent trading can cause a serious bias regarding the results of market efficiency (Miller et al. 1994). Anonymity in securities trading is at the very core of market design to promote liquidity and attract trading volume and has been rapidly growing (e.g., Comerton-Forde et al. 2018). Therefore, we made final adjustments to the model and added thin and anonymous trading.

Since we do not solely focus on a single market, it is important to take changing market dynamics (institutional, political, legal, and macroeconomic variables) into consideration (Nomran and Haroon, 2021). Following the recommendations of Ashraf (2020), we allow our model to control for country-level control variables, e.g., investment freedom, uncertainty avoidance, and GDP. There is always a probability of a spread between market risk and portfolio risk; therefore, we also control for stock-specific idiosyncratic risk in our model.

The empirical results based on the time period 2001–2020 revealed that market microstructure plays a vital role in improving the efficiency of emerging markets. The results from multivariate regression confirm that the improvement in efficiency is robust even after controlling for country-specific variables (economic growth, investment freedom, and uncertainty avoidance) and stock-specific idiosyncratic risk. The results are also robust for the alternative use of proxies of liquidity, trading cost, and volatility. This implies that both individual and institutional investors need to consider the level of market microstructure while investing and construct their stock portfolios accordingly.

The paper proceeds as follows: Section 2 highlights the literature review, with emphasis on the research gap. Section 3 focuses on the research methodology, and the fourth section discusses the major results of the study. Section 5 concludes the study.

Literature Review

2.1 Informational Efficiency

According to Fama (1970), a market is efficient if the current price reflects all the available information timely which makes it difficult to beat the market regularly. Information efficiency is important because it improves price discovery and liquidity. Literature suggests various tests to investigate informational efficiency, for example, Lo and Mackinlay (1988). These tests are based on the existence of autocorrelation in stock returns for testing the EMH. These tests have widespread uses e.g., testing the martingale hypothesis in financial time series. However, for small samples, the above-mentioned tests may produce unrealistic results. A financial time series typically exhibits heteroscedasticity. Lo and Mackinlay (1988) have insufficient properties to handle this issue as it requires ad hoc choices for selecting a lag length. Furthermore, conventional VR tests are asymptotic and the ad hoc choice of holding period limits its applicability to small samples.

2.2 Liquidity

Liquidity is one of the fundamental attributes of a well-functioning stock market (Amihud et al. 2006 and Benson et al. 2015). A highly liquid market boosts local and foreign investors confidence. In such a market, investors can engage in a large number of transactions without incurring significant transaction costs (Chordia et al. 2008). Higher trading volume reflects quick price adjustments and new information. Liquidity has the potential to improve the information flow within financial assets while promoting market efficiency (Chordia et al. 2008). The lagged shocks of price changes are found to have strong and significant effects on liquidity. It becomes quickly evident during the COVID-19 outbreak that the correlation between liquidity changes and price changes is conditioned on past information.

The increase in the compensation of discretionary liquidity traders promotes information production and efficiency. Individual investors provide liquidity and help those with private information to correct prices. Empirical support for a positive relationship between liquidity and market efficiency has been documented by Chordia et al. (2008). Frequent trading provides an opportunity for new information to be absorbed during the price discovery process. In addition, a liquid market enables market members to act on any available arbitrage opportunities, thereby making prices more efficient (Foucault et al. 2013), whereas in an illiquid market, investors will not be able to take advantage of mispricing, thus making the market less efficient (Chordia et al. 2008).

Liquidity can be assessed with different measures, e.g., stock turnover (Iwanaga and Hirose, 2022), Amivest liquidity (Díaz and Escribano, 2022), and bid-ask spread (Reichenbacher and Schuster, 2022). Each proxy captures a different dimension of liquidity, albeit with its own limitations (Goyenko et al. 2009). The selection of an appropriate liquidity proxy is an important issue in empirical research design (Goyenko et al. 2009). Therefore, unlike prior literature, we incorporate two key measures of liquidity in the present study (stock turnover and Amivest liquidity). Stock turnover measures the number of times a share is traded (Boussetta, 2022). Amivest liquidity is a dynamic measure that relates to the depth of the market by measuring the trading volume associated with a unit change in stock price. This ratio shows how well a stock can absorb trading volumes regardless of a significant price change. A high ratio shows more transactions with little effect on prices (Zhang, 2010). The case is more relevant for emerging markets such as Pakistan, China, India, Turkey, and Malaysia, as these

markets have introduced serious reforms to improve market liquidity (Arjoon, 2016). Based on the above discussion, we hypothesize that liquidity promotes informational efficiency.

2.3 Volatility

Trading decision are directly associated with future expectations, which are shaped by the arrival of new information. Its effect is reflected in stock prices, which move from equilibrium to equilibrium. Therefore, volatility is very important in the context of market efficiency and indicates market activity. The relationship, however, depends on the intensity of volatility. A period of high market volatility is often associated with lower informational efficiency (Raza et al. 2022). During such periods, a large group of investors are often engaged in noise trading, and uninformed investors make trading decisions based on fads. Hence, prices are partially driven by speculation and do not reflect firm fundamentals (Raza et al., 2022). Stable markets minimize speculative trading and enhance information incorporation into the trading decision. This results in improved efficiency, at least in its weak form.

Due to the slightly volatile nature of emerging markets makes them more relevant for the investigation of efficiency. Markets with poor governance are highly volatile and can deter informational efficiency. Low volatility is not always a sign of stability. For example, a lack of market activity in emerging markets may result in very low volatility. In such a case, an increase in volatility can improve market efficiency up to a level where volatility is not driven by market noise. Therefore, we hypothesize a very strong relationship between volatility and informational efficiency. While there is not enough market activity (e.g., the case of the PSX and KLSE) that can drive volatility above the quantity warranted by fundamentals. Hence, an increase in volatility signals can increase market activity justified by fundamentals, which should lead to market informational efficiency (Arjoon, 2016). On the basis of the theoretical and empirical justifications above, we hypothesize that excessive volatility causes informational efficiency to decline.

2.4 Trading Cost

Trading costs negatively affect investors' net returns (Boudt et al. 2019). There are several costs associated with trading, and the magnitude changes over time. For instance, over the past few decades, advancements in technology have decreased trading costs. Furthermore, transaction costs constrain trading and limit arbitrage activity, thus negatively affecting market efficiency (Chordia et al. 2008). Given that stock markets are primarily responsible for generating and combining information disseminated, it is generally important to investigate whether changes in cost levels affect how well or poorly financial markets acquire and process information (Boehmer & Kelley, 2009).

The transaction cost is directly associated with the market's ability to incorporate newly arrived information into stock prices. Raising transaction costs inevitably lessens the amount of information-based trading and reduce the informational content of stock prices (Pagano and Roell, 1996). It is thus crucial for individual and institutional investors to precisely assess the influence of costs on information gathering and processing. Trading costs reduce information acquisition and cause price informativeness (Davila and Parlato, 2021). Investor involvement in trading depends on the cost-benefit analysis of gathering information and trading. Investors' reluctance to participate in costly trading delays price reactions and lowers informational efficiency (Pagano and Roell, 1996). According to studies by Christie and Huang (1994) and Lee et al. (1993), the effective spread is a more precise indicator of a trading cost. The percentage effective and quoted spread are used to measure trading costs (Goyenko et al. 2009). Conceptually, the effective bid-ask spread measures the cost of immediate execution and is one of the most prevalent measures used in diverse applications. In addition, the effective bid-ask spread has regulatory status under Rule 605 of the US Security and Exchange Regulations (NMS), which requires all exchanges to publish their execution costs on a

monthly basis. Moreover, percentage quoted spreads require simpler datasets and are easy to calculate (Hendershot et al. 2011). Based on the rationale above, we hypothesize that lower informational efficiency is driven by higher trading costs.

2.5 Tick Size

Tick size is the minimum permissible price change in the stock price, which is determined by the exchange on which the security trades. Tick size helps improve liquidity and market quality (Zhao and Chung, 2006). A smaller tick size encourages price competition, as small price increments enable traders to express their preferences more precisely (Comerton-Forde, 2018). Due to its importance, market regulators pay increasing attention to tick size. A smaller tick size allows for tighter bid-ask spreads and increased liquidity which is why the NYSE moved from fractional to decimal pricing on January 29, 2001. Since then, the impact of decimalization on market quality has been of great interest to researchers and practitioners. Reduced tick sizes also facilitate price discovery and reduce transaction costs. The impact of tick size on informational efficiency has been the subject of substantial academic research (Zhao and Chung, 2006; Chordia et al. 2008).

Studies have shown that tick size affects market quality measures such as bid-ask spreads (Zhao and Chung, 2006), trading costs (Christie and Huang, 1994), and order book depth (Griffith and Roseman, 2019). Several studies have examined the link between tick size and price efficiency. Smaller tick sizes enhance price efficiency by narrowing bid-ask spreads and facilitating the incorporation of new information into stock prices more quickly (Chung and Chuwonganant, 2022). Tick size also has a direct impact on liquidity and trading volume (Griffith and Roseman, 2019). Higher trading volume can improve informational efficiency by increasing the availability of trading opportunities and reducing the influence of noise traders (Goldstein and Kavajecz, 2000). Tick size influences market quality, which in turn affects investor behavior. A narrower tick size can attract more market participants, including individual investors, leading to improved informational efficiency (Chung et al. 2020) and a more diverse trading environment (Goldstein and Kavajecz, 2000).

The research conducted by Yang et al. (2022) measures tick size via a uniform and stepwise system. Following Maruyama et al. (2020), we create a dummy variable with a value of 0 prior to the implementation of reducing tick size in their concerned stock markets and a value of 1 afterwards. Using this measure has the advantage of being simple, efficacious, and providing accurate results. We hypothesize that a decrease in tick size will raise the level of efficiency.

2.6 Thin Trading

Thin trading shows a period of time that is characterized by a small number of trades and it leads to low liquidity and high prices. This has a negative effect on the balance between supply and demand, creating a substantial impact on prices. The relatively lower amount of the bid-ask quote makes transactions very difficult for potential buyers and sellers. Though the overall volume is low, individual transactions tend to be large (Fizaine, 2015). That means price movements are more pronounced. In this situation, the spreads between bid and ask prices for an asset tend to be wider (Madhavan, 2000). A number of studies have suggested ways to correct for thin trading. The Kalman filter can be used to estimate the distribution of the true index. Miller et al. (1994) propose to remove the effects of thin trading by using moving averages, which reflect the number of non-trading days and adjust returns accordingly.

The absence of thin trading in any model related to market microstructure can affect the study results by posing a few potential biases (Madhavan, 2000). Prices recorded at the end of a time period have a tendency to represent the outcome of a transaction that occurred earlier in, or prior to the period in question. Thus, thin trading reduces autocorrelation in the time series of returns for a series that would

otherwise exhibit serial independence (Chordia et al. 2008). Thinly traded securities are those that cannot be traded easily without a significant change in price. They are exchanged in low volumes, often have limited numbers of interested traders, and affect market efficiency. Prior studies have tested the effect of thin trading on market efficiency; see, for example, Harrison & Moore (2012). Thin trading can be measured in many ways. Thin trading can be measured by the trading infrequency of stocks. Mobarek et al. (2008) use the methodology of Miller et al. (1994) and suggest fitting an AR(1) model to investigate thin trading. We hypothesize that thin trading hinders informational efficiency in light of the details provided above.

2.7 Anonymous Trading

Anonymity is a process in which the identities of traders are kept confidential during the execution of a transaction. This type of trading can occur on various platforms, such as dark pools or anonymous order-matching systems. Anonymous trading can attract huge investors by not revealing their identities. It affects market liquidity and overall market quality (Foucault et al. 2013). On the contrary, it may fragment liquidity across different platforms, potentially reducing overall market transparency and hindering price discovery (Chang et al. 2018). The influence of anonymous trading on market quality depends on the specific market structure and regulatory environment (Madhavan, 2000).

Studies have examined the impact of anonymous trading on price efficiency. The findings are inconclusive because some research studies suggest that anonymous trading can improve price efficiency by reducing the influence of noise traders and enhancing execution quality (i.e., Madhavan, 2000). Chang et al. (2018) conduct a natural experiment and use the same multivariate methodology to include markets in Paris, Tokyo, and Korea. They found higher liquidity in markets with anonymous trading. Anonymous trading leads to an increase in the bid-ask spread, volatility, and trading volume. Hence, the above discussion leads us to conclude that anonymity plays a vital role in improving liquidity and market efficiency. In light of the above, we hypothesize that anonymous trading increases efficiency.

2.8 Automation

Automation means replacing the conventional floor trading system with a computerized system. Under this system, trading is continuous and bids/offers are recorded and matched via an electronic database. Automation improves transaction execution speed and enhances market activity and efficiency. It facilitates wider and faster dissemination of information based on bid and offer quotes, trade sizes, and settlement prices (Hendershott and Moulton, 2011). Investors are likely to capitalize on this information when making trading decisions (Arjoon, 2016). Automation of the market lowers transaction costs, enabling more effective distribution of financial assets among investors and enhancing risk-sharing and price discovery processes (Acharya and Pedersen, 2005; Boehmer and Kelley, 2009). Freund and Pagano (2000) discussed the mechanism of automated trading systems and their merits and demerits in the context of NYSE efficiency and found a positive impact. Anderson and Vahid (2001) investigated its impact on the London and Australian stock markets using "smooth transition error correction models". They reported that futures and spot markets became more efficient post-automation. On the basis of the evidence above, we hypothesize that market automation enhances informational efficiency.

2.9 Investment Freedom

Investment freedom plays a vital role in improving market efficiency. Literature shows that investment freedom is closely associated with investor protection and confidence. It helps attract foreign investors and improve international diversification into emerging markets (Liu et al. 2018). A tight regulatory control system discourages international business expansion. Following Ashraf (2020), we include investment freedom as a control variable in our model.

2.10 Idiosyncratic Risk

The lack of importance given to idiosyncratic risk can be attributed to two factors. First, idiosyncratic risk is a matter of concern for investors with limited opportunities. Second, due to the irrational expectations of noise traders regarding idiosyncratic risk, empirical evidence is mostly based on systematic risk (Blenman and Wang, 2017). However, stock-specific idiosyncratic volatility is also an important determinant of price informativeness. Malkiel and Xu (2002) found a positive relationship between idiosyncratic volatility and expected returns, suggesting that idiosyncratic risk is more important. Angelidis (2010) finds idiosyncratic risk lower in 24 emerging markets due to increased efficiency.

2.11 Uncertainty Avoidance Index

Societies with high uncertainty avoidance perceive threats as more likely in uncertain situations. The degree of comfort is easily reflected in financial decisions. Higher uncertainty avoidance individuals have a delayed response to new information, which reduces informational efficiency (Hofstede, 1984). Dou et al. (2016) suggest research on cultural dimensions to explore uncertainty avoidance's impact on stock price informativeness. This highlights culture's fundamental role in shaping firm stock prices. Uncertainty avoidance measures individuals' tolerance and fear of uncertainty. It implies that stock prices in countries with higher uncertainty avoidance would exhibit lower anticipation of future earnings and inefficiency. Dou et al. (2016) show that investors embedded in cultures with higher uncertainty avoidance exhibit a lower (higher) under-reaction to earnings information. We show that uncertainty avoidance also influences investors' ability to anticipate future earnings and impound this information into current prices.

2.12 Gross Domestic Product

GDP and market efficiency are key economic concepts that shed light on the overall health and performance of an economy. The GDP-market efficiency link is intricate. Economic growth has a positive effect on almost all sectors, including stock prices and market efficiency (Chen et al. 2021). Robust GDP sustains market efficiency. High consumer demand and commercial opportunities in a rising economy might draw more investors and boost market activity. By enhancing competition, facilitating price discovery, and lowering information asymmetry, this influx of participants and resources might improve market efficiency. In this regard, strong GDP can promote market efficiency by boosting economic activity and luring investment, while efficient markets can support economic growth by efficiently allocating resources (Chen et al. 2021). These countries have different economic growth rates, and to control for variation in economic growth, we follow Ashraf et al. (2020) and Nomran and Haroon, (2021) and include GDP in our baseline model.

3. Data Collection and Methodology

This study examines listed firms in five Asian emerging markets: China, Pakistan, Malaysia, Turkey, and India. These markets are a growing force in the world economy, and their respective contributions have almost doubled in the last two decades. These markets account for 1/5th of the world exports and receive 1/3 of the FDI from emerging markets. Third, during the past three decades, these markets have proactively pursued internationalization and made considerable efforts to enhance the market microstructure (Raza et al. 2022). The final sample is selected via a stratified random sampling (equal allocation) technique, and the resultant sample is composed of 250 firms, which exceed 5% of the total population. The analysis is carried out for 2001-2020.

The informational efficiency of all individual stocks is measured over a rolling window of 24 months with the variance ratio (VR) test (Kim, 2006). This study focuses on seven key components of market microstructure: liquidity, volatility, trading costs, tick size, anonymous trading, thin trading, and automation. We use two proxy measures liquidity, volatility, and trading costs. Volatility is measured with EGARCH and TGARCH, all estimated over a rolling two-year window. For the estimation of these variables, the closing price and volume of all stocks are obtained monthly from the CRSP database and Yahoo Finance. Trading costs, tick size, and thin trading data for each market are obtained from the Trade and Quote (TAQ) database. Anonymous trading and automation data is obtained from the relevant stock markets and security exchanges.

This study prefers the panel data analysis technique over other panel models and classical event study methodologies due to several reasons: Firstly, market microstructure and efficiency evolve over time, and it's not a momentary event (Laogarde-Segot, 2009). Second, panel data regression is better at capturing the time varying relationship between dependent and independent variables (Ashraf, 2020). Third, panel data analysis extracts both cross-sectional and time series variation from the underlying panel data and counts for Multicollinearity, Heteroscedasticity, and estimation bias (Woolridge, 2010). To bridge the link between market microstructure and information efficiency, we specify the following baseline model:

$$IE_{it} = \alpha_i + \psi_t + \beta_1 LIQ_{it} + \beta_2 VOL_{it} + \beta_3 TC_{it} + \beta_4 TS_{it} + \beta_5 TT_{it} + \beta_6 AUT_{it} + \beta_7 ANT_{it} + \beta_8 GDP_{it} + \beta_9 SML_{it} + \beta_{10} UA_{it} + \beta_{11} IDRSK_{it} + \epsilon_{it} \quad (1)$$

In the above equation, IE is the dependent variable (informational efficiency) of a particular firm i at time t . α_i is the individual stock effect and β is the beta coefficient of each independent variable. The ψ is time effect. LIQ is liquidity, VOL is volatility, TC is trading cost, TS is tick size, TT is thin trading, AUT is automation, ANT is anonymous trading, GDP is gross domestic product, SML is stock market liberalization, UA is uncertainty avoidance, and $IDRSK$ is idiosyncratic risk, respectively. The uncertainty avoidance index is taken from Hofstede et al. (2010)'s framework of national culture and measures cross-country differences in the level of uncertainty aversion among investors. The data on investment freedom is collected from the Heritage Foundation database of economic freedom. The historical GDP is collected from the World Bank (WDI).

Variable Measures

3.1 Informational Efficiency

Informational efficiency is assessed with the VR test (Kim, 2006; Arjoon, 2016; and Said et al. 2022). This test accounts for autocorrelation among security prices, identifies random walk, and shows the magnitude of informational efficiency. The key reason for using the VR test is its widespread use in the literature and robustness in small samples of panel data. The application of wild bootstrap enhances its robustness to size distortions and leads to a substantially higher power than other VR tests. This test is also applicable to data with an unknown form of conditional and unconditional heteroscedasticity.

This study follows the methodology of Kim (2006) and applies the wild bootstrap to the VR tests of Lo and Mackinlay (1998). The wild bootstrap test built on it is implemented in the following three steps:

- Form a bootstrap sample of T observations $X_t^* = \eta_t X_t$ ($t = 1 \dots T$) where η_t is a random sequence with $E(\eta) = 0$ and $E(\eta^2) = 1$.
- Calculate $MV^* = MV_2(X^*; k_i)$, the $MV_2(X^*; k_i)$ statistic which is acquired from the bootstrap sample created in stage first.
- Repeat stage (a) and stage (b) passably 'm' times to make a bootstrap distribution of the test statistic $\{MV_2(X^*; k_j; j)\}_{i=1}^m$.

3.2 Volatility

In the main analysis, we measured volatility with the EGARCH model. This model captures several features of volatility in stock returns, e.g., fat tails and volatility clusters. We used the Exponential GARCH model developed by Nelson (1991). An EGARCH model is expressed as follows.

$$\ln(\sigma_t^2) = \phi + \psi \ln(\sigma_{t-1}^2) + \Upsilon \frac{\epsilon_{t-1}}{\sqrt{\sigma_{t-1}^2}} + \eta \left[\frac{|\epsilon_{t-1}|}{\sqrt{\sigma_{t-1}^2}} - \sqrt{\frac{2}{\pi}} \right] \quad (2)$$

In the above eq. (2), σ_t is the conditional variance at time t , ϕ is constant, Υ is an asymmetric effect, ψ is a GARCH effect and ϵ_t are the residual errors. In the robustness analysis, we use an alternative measure (TGARCH) for the measurement of volatility. As discussed by (Arjoon, 2016) this model can capture the asymmetric leverage effects (effects of good and bad news) on the level of volatility. This model is specified below.

$$R_t = \alpha + \beta R_{t-1} + \epsilon_t \quad (3)$$

$$VOL_{ij} = \frac{1}{D_{ij}} \sum_{t=1}^{D_{ij}} \sigma_{ijt}^2 \quad (4)$$

In eq. (4), VOL_{ij} is the volatility of stock i in year j , while σ_{ijt}^2 is the daily fitted value of σ^2 of stock i in year j . We assess the robustness of EGARCH using the Threshold GARCH model.

3.3 Liquidity

Literature assesses liquidity with different measures, i.e., Amivest Liquidity (Goyenko et al. 2009; Díaz & Escribano, 2022), Stock Turnover (Iwanaga & Hirose, 2022), and Bid-Ask Spread (Reichenbacher & Schuster, 2022). Following Iwanaga & Hirose (2022), we assess liquidity with stock turnover. It shows liquidity, and it measures the number of times a share has been traded (Boussetta, 2022). The higher number of trades signals the incorporation of newly arrived information. Arjoon (2016) focused directly on the number of transactions to measure liquidity; however, we adopt the modification of (Chordia et al. 2008) and consider a relative measure by dividing the total number of shares traded over a period by the average number of shares outstanding. Specifically, it is measured as,

$$STO_{it} = \frac{\text{Total Number of Shares traded (Volume)}_{it}}{\text{Average Number of shares outstanding}}$$

We also use Amivest liquidity in our model. It is a dynamic measure, and it relates to the depth of the market by measuring the trading volume associated with a unit change in stock price. Limited price sensitivity is desirable as it reduces volatility and uncertainty in the marketplace. This ratio shows how

well a stock can absorb trading volumes regardless of a significant price change. A high ratio shows more transactions with little effect on prices (Zhang, 2010). It can be measured as

$$Amivest_{it} = \frac{\sum_t VOL_{it}}{\sum_t |R_{it}|}$$

VOL_{it} represent daily trading volume and $|R_{it}|$ is the daily absolute stock returns of firm i at time t . The higher the value of Amivest, the higher the firm's liquidity.

3.4 Trading Cost

The studies conducted by Lee et al. (1993) and Christie and Huang (1994) suggest that the effective spread is a more accurate indicator of trading costs than the quoted spread. We adjust quotes by five seconds in accordance with Lee et al. (1993). To examine trading cost the percentage effective spread, and depths are calculated as follows:

$$Percentage\ Effective\ Spread_{it} = \frac{100(|Price_{it} - Ask_{it} + Bid_{it}|/2)}{((Ask_{it} + Bid_{it})/2)}$$

For robustness, we use percentage quoted spread as follows:

$$Percentage\ Quoted\ Spread = \frac{(A_{it} - B_{it})}{(A_{it} + B_{it})/2}$$

A_{it} and B_{it} represent the best bid and ask prices at Trade and Quote (TAQ) database for the listed firm i at time t , respectively.

3.5 Thin Trading, Anonymous trading and Automation

This study used "Bid-Ask Spread" to measure thin trading. A lower spread facilitates traders' execution of transactions. A large spread constrains investors' participation and impedes efficiency (Hendershott & Moulton, 2011). Firm-level anonymous trading data is obtained either from the firm's website or from relevant stock exchanges. Anonymous trading magnitude is measured by the disclosure score. Each firm receives points for the number of anonymous trades they publish on their websites. The anonymous trading score is reflected as an index where, at each point in the period, the score is aggregated for each firm. If a firm holds more anonymous trades, it will have a higher score. We measure automation using a dummy variable with a value of 0 prior to automated trading introduction and a value of 1 afterwards.

Controlling variables

This study uses four controlling variables in the model as suggested by the previous literature (i.e., Ashraf, 2020). These are INVFF, UA, IDRSK, and GDP. We evaluate investment freedom with the INVFF Index developed by the Heritage_Foundation (2018)¹, which ranks each country on a score of 1–100. It evaluates country wise restrictions imposed on investments. A country with an attractive (conservative) investment environment is ranked high (lower) on the index. For uncertainty avoidance, this study uses the uncertainty avoidance index (UAI) developed by Hofstede (1984). The index ranks each country on a score of 1–120. Following Fu (2009) methodology, the idiosyncratic risk is calculated by first running the Fama-French three factor model and obtaining the residual from that model. Then, using the 3-factor Fama-French model, we calculated the standard deviation of these residuals, which reflect the idiosyncratic risk three-factor regressions for each firm in each year.

¹ <https://www.heritage.org/international-economics/commentary/2018-index-economic-freedom>

4. Results and Interpretations

We initiate the analysis by reporting descriptive statistics in Table 1. The basic purpose of reporting these statistics is to highlight the general characteristics of all variables for the time period 2001–2020. Table 1 shows that all variables across all markets have varying descriptive characteristics. For a comparative analysis of the level of informational efficiency across the five markets, let us zoom in on the first column of Table 1.

We can see that among all the five markets, SSE achieves the highest level of informational efficiency. It outperforms PSX, KLSE, NSEI, and BSI by 13, 60, 17, and 27 basis points, respectively. Among all markets, the SSE is the most liquid but also the most risky, followed by the Indian stock market. The results are a bit different when alternative proxies are used to assess volatility. Trading costs are highest in NSEI, which is one of the second-largest performing developing Asian economies after SSE, followed by PSX. Stock markets frequently engage in anonymous trading, which results in moderate fluctuations across emerging countries. Further, the dispersion of informational efficiency and microstructure components is positively skewed, while the control variables are negatively skewed. On the contrary, the kurtosis results suggest that efficiency and market microstructure variables are highly peaked (excessive leptokurtic). In other words, these variables for the given period can be described as asymmetric. These results are consistent with the Jarque-Bera normality test.

Table 1²: General characteristics of all dependent and explanatory variables across the five markets.

	IE	AMVST	STO	EGARCH	TGARCH	TC1	TC2	TS	TT	ANT	AUT	GDP	SML	UA	Id. Risk
Panel I Summary statistics of 50 firms listed on the Pakistan Stock Exchange (PSX).															
μ	1.31	0.57	25.39	0.03	0.04	20.52	0.057	0.090	14.10	0.35	0.65	26.17	45.52	35.50	-0.19
σ	0.24	1.92	33.34	0.07	0.06	14.25	0.113	0.031	35.30	0.47	0.39	0.21	8.57	21.57	1.76
Sk.	-0.15	7.77	29.02	7.71	6.30	4.34	5.710	0.20	30.75	0.62	-1.02	-0.10	-0.47	-1.06	6.18
Kurt	-0.18	89.38	87.62	39.51	43.82	30.28	55.62	-2.18	94.09	1.71	1.05	2.03	1.82	1.79	99.96
JB	33.25	30.49	30.13	62.41	72.31	50.22	35.52	54.7	35.43	37.91	47.81	38.38	90.42	65.05	37.82
No's	3151	3151	3151	3151	3151	3151	3151	3151	3151	3151	3151	3151	3151	3151	3151
Panel II Summary statistics of 50 firms listed on the Kuala Lumpur Stock Exchange (KLSE).															
μ	1.78	13.39	92.52	0.09	0.06	7.65	0.02	0.087	0.03	0.20	0.57	26.25	45.29	18.51	-10.63
σ	0.20	0.22	20.15	0.30	0.11	9.92	0.04	0.03	0.20	0.42	0.36	0.24	11.84	10.68	7.04
Sk.	-0.13	9.93	4.215	7.13	2.11	1.75	5.90	0.40	5.23	1.5	-0.42	-0.06	-0.01	-0.00	-0.42
Kurt	-0.90	9.00	25.89	61.81	5.98	4.51	10.61	-2.01	57.40	0.69	1.45	1.76	1.49	1.79	1.81
JB	30.02	0.72	2295	13.71	10.05	15.50	11.05	49.3	11.49	43.8	42.6	57.50	85.25	54.47	86.31
No's	3151	3151	3151	3151	3151	3151	3151	3151	3151	3151	3151	3151	3151	3151	3151
Panel III Summary statistics of 50 firms listed on the Shanghai Stock Exchange (SSE).															
μ	1.18	1.31	1.88	0.24	0.35	8.75	0.03	0.075	0.04	0.40	0.62	29.76	26.47	15.50	-25.03
σ	0.29	4.16	1.83	0.03	0.02	8.91	0.05	0.02	0.08	0.48	0.36	0.40	3.74	8.88	29.20
Sk.	0.39	6.48	14.62	12.62	7.85	4.55	10.15	1.5	5.64	0.40	-0.94	-0.34	-0.53	-2.77	-0.69
Kurt	0.11	4.96	26.57	45.71	89.67	15.65	23.05	0.69	45.80	2.01	1.82	1.91	1.95	1.79	2.08
JB	28.35	0.15	25.46	13.79	27.48	21.32	22.35	29.3	69.41	46.2	620	57.99	79.70	51.71	98.20
No's	3151	3151	3151	3151	3151	3151	3151	3151	3151	3151	3151	3151	3151	3151	3151
Panel IV Summary statistics of 50 firms listed on the National Stock Exchange (NSEI).															
μ	1.28	0.21	10.35	0.19	0.21	40.25	0.16	0.081	2.35	0.22	0.87	27.50	35.31	19.05	-09.3
σ	0.55	0.15	2.25	0.17	0.14	35.06	0.14	0.028	0.17	0.41	0.25	0.37	7.02	12.05	8.03
Sk.	0.65	1.85	-0.14	3.95	3.15	1.29	2.51	0.87	6.02	1.50	-2.55	0.02	0.39	0.042	0.54
Kurt	3.07	5.23	2.05	35.7	21.8	2.75	11.20	-1.24	29.21	0.69	8.05	1.55	2.25	1.82	1.73
JB	33.41	45.01	29.05	39.1	45.4	26.04	34.51	36.5	52.03	38.9	43.01	47.09	51.03	42.02	78.12
No's	3151	3151	3151	3151	3151	3151	3151	3151	3151	3151	3151	3151	3151	3151	3151
Panel V Summary statistics of 50 firms listed on the Istanbul Stock Exchange (BSI).															
μ	1.45	0.54	13.80	0.151	0.171	0.265	0.014	0.10	0.12	0.25	N/A	28.01	60.6	42.05	-0.24
σ	0.54	1.50	2.95	0.08	0.09	0.29	0.88	0.02	0.27	0.43	N/A	0.321	8.95	25.06	3.06
Sk.	0.75	6.75	-0.82	1.92	6.54	2.55	105.1	-0.62	3.21	1.15	N/A	-0.06	-0.13	-0.08	3.14
Kurt	3.25	52.5	47.1	8.09	85.21	9.65	9.72	-1.71	72.05	-0.49	N/A	-1.27	-1.62	-1.56	72.06
JB	125.3	145.1	165.8	328.5	423.8	715.2	815.3	48.4	112.9	51.4	N/A	978.1	155.4	155.4	43.02
No's	3151	3151	3151	3151	3151	3151	3151	3151	3151	3151	N/A	3151	3151	3151	3151

The above statistics are for the whole sample firm across the whole time period. It provides us with an overview of the general trend. However, such analysis fails to grasp the changing dynamics in

² The above table 1 presents summary statistics for all five stock markets (PSX, KLSE, BSI, NSEI, and SSE) for 2001-2020. The symbol IE represents informational efficiency. AMVST and STO represent Amivest and Stock Turnover. TC, TS, TT, AUT, and ANT show transaction cost, tick size, thin trading, automation, and anonymous trading. Percentage quoted spread and percentage effective spread measure trading cost (TC1 and TC2).

informational efficiency across firms. For this purpose, we plot the individual values of a few variables of the top 50 firms in each market in Figure 1.

Over the past two decades, the top 05 firms in the sample markets are shown in Fig. I (volatility) and II (liquidity). SSE has the highest volatility, while KLSE has the lowest. Some firms have high volatility (between 0.07 and 0.04), while the majority have moderate volatility (between 0.03 and 0.02). SSE liquidity is the highest, followed by NSEI and BSI. KLSE and PSX are among the least liquid markets.

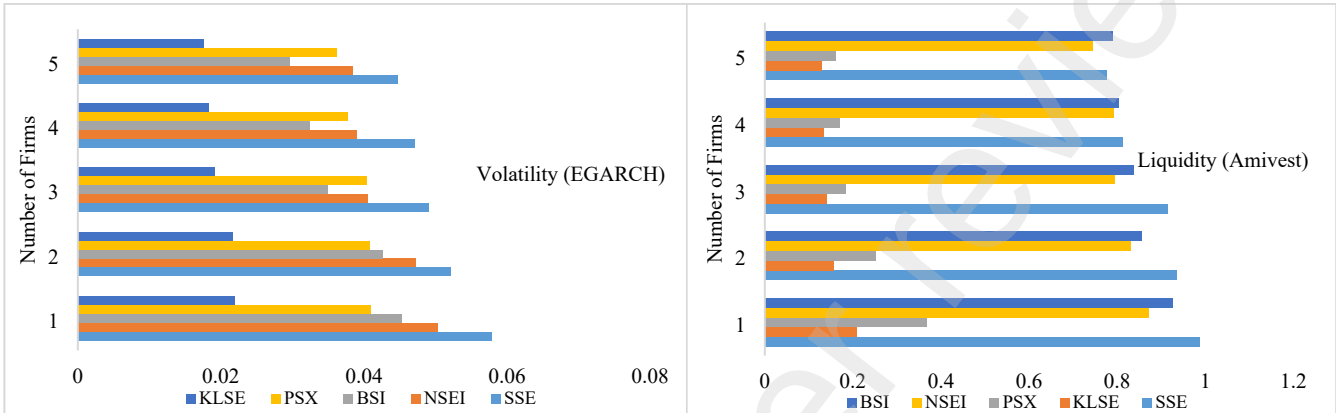


Figure. I Volatility of Asian Emerging Markets (2001-2020).

Figure. II Liquidity of Asian Emerging Markets (2001-2020).

4.1 Correlation Analysis

Table 2 reports a pairwise correlation between informational efficiency and the rest of the variables. The results show that the battery of variables used for the main analysis and robustness analysis are generally consistent. Recall that a higher (lower) value for informational efficiency indicates a lower (higher) efficiency. It is important because a negative correlation coefficient in this case does not show a decrease in informational efficiency. Instead it implies that the estimated value of the VAR test will decrease which indicates improvement in informational efficiency. For example, the correlation coefficients for both liquidity measures are negative which indicates a positive association between liquidity and informational efficiency.

The correlation coefficients in these markets for AMVIST and STO range from -0.20 and -0.50 respectively. These results show a positive association between liquidity and informational efficiency. Albeit, the magnitude of association is weak in the majority of cases. We didn't notice any significant deviation from these results in the rest of the markets and the choice of proxy used. In order to obtain the difference in the strength of association across markets we also observed that thin trading asserts that in this situation there are very few traders. Therefore, a few traders are insufficient to represent the entire market and show a negative association with market efficiency. The case of automation shows a significant association, and SSE has a high association, confirming that when markets automated, efficiency raises. Trading cost results are consistent across markets but particularly highly significant in SSE. An inverse association implies that an increase in trading cost is more likely to move away from fundamental values and join noise trading which leads to lower efficiency. There is an inverse relationship between trading cost and efficiency, demonstrating that trading cost reduces the informativeness of stock prices.

In SSE, NSEI, and BSI, anonymous trading has a high magnitude of association with efficiency. These markets are leading markets and handle large trades. Market efficiency increases due to high trading volumes. Liquidity and efficiency are also increased by large anonymous trading. Tick size exhibits significant but weak magnitude due to low concern and gradual decrease. This shows only a slight association with efficiency while highly correlated with trading cost as trading costs significantly decrease with a decrease in tick size. The results reveal a negative correlation between tick size and efficiency and liquidity, while a positive correlation between trading costs. As tick size decreases, liquidity and efficiency increase.

Volatility measures, except SSE and BSI, are negatively and highly correlated among the whole battery of variables with efficiency. This suggests that excessive volatility reduces efficiency. Another major finding is that liquidity and volatility are positively and weakly correlated in PSX, KLSE, and NSEI. The reason is that these markets are still in their early stages of development. This implies that activity magnitude is not significant enough to affect efficiency. These markets lack excess volatility, which causes lower liquidity, while there is a negative and weak correlation between liquidity and volatility (in SSE and BSI). A relatively illiquid market places orders and affects the market's capacity to absorb shocks. Therefore, investors find it relatively easy to push prices up and down. High volatility can distort resource allocation by making investors reluctant to hold stocks. Risk-averse investors demand a high-risk premium, which increases costs and reduces market liquidity. Lower liquidity generally results in a more volatile market and causes security prices to change more frequently; whereas high liquidity usually creates a less volatile market in which security prices don't fluctuate significantly. The results also highlight the importance of the controlling variables. The correlation coefficients for INVF, GDP, IDRSK, and UA show a weak to moderate association with informational efficiency albeit the results are insignificant in some cases.

Table 2:³ Correlation Analysis of Informational Efficiency, Market Microstructure with Controlling Variables

Panel A	Correlation Analysis of Firms Listed on the Pakistan Stock Exchange (PSX)													
	Inf-Eff	Amivest	STO	TT	AUT	TC1	TC2	ANT	TS	EGARCH	TGARCH	SML	GDP	IDRSK
AMVST	-0.27**													
STO	-0.32*	0.14***												
TT	0.15**	-0.16***	-0.10											
AUT	-0.19*	0.29	0.31	0.05										
TC1	0.29**	-0.06*	-0.11	0.01**	-0.01*									
TC2	0.23*	-0.02	-0.24	0.36*	-0.23*	0.35**								
ANT	-0.22*	0.21**	0.24*	-0.04	0.02	-0.03*	-0.05*							
TS	0.14*	-0.11**	-0.13*	-0.02	0.02	-0.33**	-0.41*	0.08						
EGARCH	-0.47**	0.33*	0.26*	0.13**	0.31**	-0.21*	-0.22**	0.38**	-0.12**					
TGARCH	-0.51**	0.37*	0.23**	0.11*	0.42*	-0.16**	-0.23*	0.37**	-0.14*	0.25***				
SML	-0.09	0.11	0.12	-0.21**	0.21	-0.22	-0.08**	0.06*	-0.02	0.22**	-0.03			
GDP	-0.13*	0.10	0.07	0.18**	0.07***	-0.25	-0.16***	0.25	0.20	-0.39***	-0.05	0.33***		
IDRSK	-0.29***	0.19***	-0.07**	-0.043	-0.242	0.01	0.01	0.01*	-0.01	0.04	0.12***	-0.03	0.24	
UA	0.15	-0.41	-0.32	0.17**	-0.18	-0.19***	0.04	0.08	-0.09	-0.40**	-0.05*	0.32***	0.64***	0.10
Panel B	Correlation Analysis of Firms Listed on the Kuala Lumpur Stock Exchange (KLSE)													
	Inf-Eff	Amivest	STO	TT	AUT	TC1	TC2	ANT	TS	EGARCH	TGARCH	SML	GDP	IDRSK
AMVST	-0.31**													
STO	-0.48**	0.10												
TT	0.13**	-0.11*	-0.07**											
AUT	-0.16**	-0.16***	-0.07	-0.04*										
TC1	0.33**	-0.17**	-0.02	0.08*	-0.05*									
TC2	0.36*	0.15**	-0.11***	0.27**	-0.06	-0.01								
ANT	-0.26*	0.10*	0.13*	-0.15	0.29	-0.29*	-0.35*							

³ To assess the association between the study variables targeting the period 2001-2020, Pearson correlation analysis is presented in Table 2. Inf-Eff is informational efficiency of the firm, AMVST and STO are stocks liquidity, TT is thin trading, ANT is anonymous trading, Volatility is measured with EGARCH and TGARCH, TC1 and TC2 is trading cost, AUT is automation, TS is tick size, IDRSK is idiosyncratic risk, GDP is concern country Gross Domestic Product, UAI is uncertainty avoidance index, and SML is stock market liberalization.

TS	0.12*	-0.01**	-0.03*	-0.03	0.01	-0.42*	-0.48	0.04							
EGARCH	-0.42**	0.23**	0.22*	0.17	0.08	-0.31*	-0.35**	0.07**	-0.03*						
TGARCH	-0.43*	0.18	0.26	0.19	0.11**	0.23*	-0.26**	0.09**	-0.07*	0.49***					
SML	-0.23	0.14	0.27	-0.09	0.64***	-0.05	-0.13	0.08	-0.03*	0.01*	0.31				
GDP	-0.34*	0.21	0.23	-0.05	0.67**	-0.05***	-0.02	0.06	0.02	-0.05	0.27	0.95***			
IDRSK	-0.32*	0.06**	-0.07*	0.13	-0.10*	0.08	0.10	0.17	0.16	0.43*	0.40	-0.51**	0.34*		
UA	0.24	-0.01	-0.03	-0.05	0.65**	-0.05	-0.03	0.01	-0.42	0.01	0.02	0.94***	0.99***	0.39***	
Panel C Correlation Analysis of Firms Listed on Shanghai Stock Exchange (SSI)															
	Inf-Eff	Amivest	STO	TT	AUT	TC1	TC2	ANT	TS	EGARCH	TGARCH	SML	GDP	IDRSK	
AMVST	-0.35***														
STO	-0.45**	0.28***													
TT	0.35***	-0.13*	-0.032												
AUT	-0.33**	0.01	0.15**	0.03											
TC1	0.27***	-0.13***	0.01	0.06***	-0.12***										
TC2	0.24**	-0.04*	-0.03	0.29*	-0.14**										
ANT	-0.39**	0.20**	0.26*	-0.16*	0.04	-0.15**	-0.24*								
TS	0.18**	-0.13***	-0.14*	-0.17	0.04	-0.39**	-0.41*	0.25							
EGARCH	0.58***	-0.26*	-0.20*	-0.25**	0.17**	-0.35**	-0.35**	0.24**	-0.41**						
TGARCH	0.62**	0.31	-0.23*	0.31*	0.13**	0.37**	0.39*	0.29*	-0.05*	0.90***					
SML	-0.09	0.02	0.04	-0.06**	0.26**	0.11***	0.05*	0.05	-0.09*	0.07	0.04				
GDP	-0.14**	0.01	0.06**	0.00	0.63**	-0.15**	-0.08**	0.04	-0.07*	-0.03	0.19	-0.60***			
IDRSK	-0.24***	0.03	-0.04*	-0.29**	-0.04*	0.03	0.02	0.24	0.04	0.08**	0.05*	0.34**	0.04		
UA	0.03	0.01	0.07**	-0.00	0.57***	-0.15***	-0.09***	0.06	-0.16	0.02	0.10	-0.60***	0.99***	0.39***	
Panel D Correlations Analysis of Firms Listed on National Stock Exchange of India (NSEI)															
	Inf-Eff	Amivest	STO	TT	AUT	TC1	TC2	ANT	TS	EGARCH	TGARCH	SML	GDP	IDRSK	
AMVST	-0.22**														
STO	-0.25*	0.23***													
TT	0.19**	-0.08*	-0.04												
AUT	-0.10***	0.09*	0.06*	0.09											
TC1	0.23**	-0.18***	-0.15*	0.20***	-0.32*										
TC2	0.33*	-0.16***	-0.21**	0.24**	-0.24*	0.46***									
ANT	-0.37**	0.36**	0.39*	-0.05*	0.07	-0.12**	-0.17*								
TS	0.17**	-0.24**	-0.23	-0.04	0.08	-0.46**	-0.52*	0.05							
EGARCH	-0.47**	0.21*	0.31**	-0.25*	0.04*	-0.27**	-0.21	0.28**	-0.39*						
TGARCH	-0.45**	0.22*	0.34*	-0.16*	0.02	0.35*	-0.16*	0.44**	-0.25*	0.33***					
SML	-0.12*	0.04	0.03	0.01	0.42***	-0.01	-0.01	0.02	-0.45*	-0.01	-0.02				
GDP	-0.14*	-0.11**	0.07	0.08	0.50***	-0.13***	0.06*	0.03	-0.05*	-0.02	-0.05*	-0.13***			
IDRSK	-0.37***	0.14	-0.06*	-0.29**	-0.01	0.03	0.09	0.33	0.06	0.07*	0.04*	0.01	0.54		
UA	-0.05*	-0.1***	0.03	-0.07	0.49***	-0.13	0.07**	0.74	-0.09	-0.02	-0.05*	-0.13***	0.67***	0.03	
Panel E Correlations Analysis of Firms Listed on Turkey Stock Exchange (BSI)															
	Inf-Eff	Amivest	STO	TT	AUT	TC1	TC2	ANT	TS	EGARCH	TGARCH	SML	GDP	IDRSK	
AMVST	-0.32**														
STO	-0.21*	0.04*													
TT	0.17**	-0.06*	-0.05												
AUT	N/A	N/A	N/A	N/A											
TC1	0.26**	-0.03	0.18**	0.26**	N/A										
TC2	0.24*	-0.13	-0.12	0.34*	N/A	-0.05*									
ANT	-0.42**	0.36**	0.39*	-0.34	N/A	-0.24*	-0.26*								
TS	0.15**	-0.22*	-0.26*	0.02	N/A	-0.36**	-0.37*	0.08							
EGARCH	0.45**	-0.44*	-0.46	-0.14**	N/A	-0.37**	-0.27	0.20**	-0.04*						
TGARCH	0.55*	-0.37	0.27*	-0.35**	N/A	-0.32*	0.36	0.32*	-0.08	0.65***					
SML	-0.52*	0.01	-0.07**	-0.03	N/A	-0.21***	0.03	0.13	-0.00**	-0.18***	-0.14***				
GDP	-0.06*	0.06**	-0.27***	-0.27	N/A	-0.20***	0.13	0.17	-0.10*	-0.20***	-0.13***	0.71***			
IDRSK	-0.33*	0.17**	-0.15**	0.45	N/A	0.04	0.05	0.38	0.03	0.08*	0.40*	0.07	0.37		
UA	-0.17**	0.25	-0.07***	-0.13	N/A	-0.19***	0.06	0.43	-0.06	-0.20***	-0.12***	0.62***	0.64***	0.07	

*, **, *** Indicates significance levels at 10%, 5% and 1%.

4.2 Effect of Market Microstructure Components on Informational Efficiency

Following Chordia et al. (2008) and Chung and Hrazdil (2010), we rely on the GMM model for panel data estimation. The model uses the 1st lag of the dependent variable with four different specifications (1-4) of the baseline model. Using OLS or fixed and random effect estimation for a dynamic panel model results in a dynamic panel bias due to the correlation between the lagged dependent variable and unobserved individual effects. Arellano and Bover (1995) and Blundell and Bond (1998) further developed the Arellano and Bond estimation methods into a method collectively known as the System Generalized Method of Movement (SGMM). By transforming the instruments, themselves, the fixed effects become exogenous to the instruments themselves. This estimator allows efficiency gains over the difference GMM estimator (Blundell and Bond, 1998).

Table 3. Results of panel data country-wise regression based on the system GMM.

Panel 1. Pakistan Stock Exchange				Panel 2. Kuala Lumpur Stock Exchange				Panel 3. Shanghai Stock Exchange				Panel 4. Indian Stock Exchange				Panel 5. Turkey Stock Exchange				
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
IE (-1)	0.25** (0.03)	0.24* (0.01)	0.24*** (0.00)	0.24*** (0.00)	0.29*** (0.00)	0.16*** (0.00)	0.12*** (0.00)	0.12*** (0.00)	0.34*** (0.00)	0.35*** (0.00)	0.34*** (0.00)	0.35*** (0.00)	0.28*** (0.00)	0.29** (0.04)	0.28* (0.07)	0.25* (0.06)	0.26** (0.03)	0.26** (0.05)	0.32** (0.04)	0.35** (0.03)
AMVST	-0.15** (0.04)	-0.16* (0.06)			-0.12** (0.05)			-0.15* (0.05)	-0.21*** (0.03)			-0.23** (0.04)	-0.17** (0.04)	-0.19* (0.08)			-0.16** (0.02)		-0.14* (0.07)	
EGARCH	-0.14*** (0.01)	-0.17** (0.03)		-0.15** (0.02)			-0.14** (0.04)		0.31*** (0.02)	0.35*** (0.01)				0.28** (0.04)			0.26** (0.05)			
TC1			0.11** (0.04)	0.10*** (0.00)	0.09** (0.03)	0.13* (0.02)			0.07** (0.04)	0.08* (0.07)			0.12* (0.09)		0.16* (0.07)			0.22* (0.06)		0.18* (0.08)
TT			0.23*** (0.01)				0.19** (0.02)				0.42*** (0.00)				0.39*** (0.00)					0.35*** (0.04)
ANT	-0.09* (0.05)			-0.08* (0.06)	-0.06** (0.05)					-0.13*** (0.00)				-0.16** (0.03)						-0.14** (0.05)
AUT		-0.09* (0.07)	-0.07* (0.06)			-0.07** (0.04)		-0.05* (0.08)		-0.10** (0.05)	-0.15* (0.07)			-0.15** (0.02)		-0.11* (0.05)	N/A	N/A	N/A	N/A
TS				0.03* (0.07)	0.02* (0.08)				0.04** (0.05)					0.06** (0.05)						0.05** (0.04)
STO				-0.14* (0.08)		-0.15* (0.02)				-0.15** (0.04)						-0.16** (0.05)		-0.17** (0.05)		
TGARCH			-0.45 (0.23)			-0.35* (0.09)			0.63 (0.37)			0.39* (0.06)				0.12* (0.07)		0.04* (0.06)		
TC2	0.17** (0.05)	0.16** (0.04)					0.15** (0.02)	0.18*** (0.00)			0.38** (0.05)	0.33* (0.08)	0.35* (0.07)			0.34* (0.06)	0.29 (0.41)		0.39 (0.29)	
IDRSK	-0.14** (0.03)	-0.20** (0.05)			-0.08* (0.06)	-0.06** (0.05)					-0.07* (0.08)	-0.03** (0.04)		-0.03** (0.03)	-0.07** (0.05)		-0.11** (0.05)		-0.16* (0.09)	
GDP		-0.27** (0.02)		-0.29** (0.04)	-0.36** (0.04)	-0.31** (0.05)				-0.26** (0.04)			-0.32** (0.05)		-0.36* (0.07)				-0.25* (0.09)	
SML	-0.08* (0.06)	-0.09 (0.13)					-0.03 (0.30)			-0.20 (0.42)	-0.16** (0.04)		-0.02* (0.08)			-0.03* (0.07)		-0.02* (0.06)		
UA	0.17** (0.01)		0.01*** (0.00)		0.51*** (0.00)		0.37*** (0.00)		0.03* (0.05)			0.02*** (0.00)		0.16* (0.07)	0.14* (0.06)				-0.01** (0.04)	
Sarg. Test	0.14	0.13	0.81	0.21	0.17	0.19	0.26	0.41	0.31	0.17	0.25	0.14	0.30	0.13	0.197	0.00	0.242	0.208	0.465	0.105
AR(1)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.000	0.391	0.000	0.000	0.000	0.000
AR(2)	0.12	0.31	0.17	0.11	0.19	0.28	0.71	0.18	0.21	0.61	0.58	0.34	0.28	0.24	0.29	0.152	0.278	0.903	0.481	0.675
No's	3151	3151	3151	3151	3151	3151	3151	3151	3151	3151	3151	3151	3151	3151	3151	3151	3151	3151	3151	3151

*, **, *** Indicates significance levels at 10%, 5% and 1%.

The study is a two-step system GMM estimation technique. This technique mitigates endogeneity bias in panel data model specifications. This originates from the dynamic link between efficiency and microstructures. Endogeneity bias can lead to inconsistent estimates and incorrect inferences, which may lead to ambiguous conclusions and incorrect theoretical interpretations. Such bias can even lead to wrong sign coefficients. Our main hypothesis is that market microstructure improves informational efficiency. However, if this is true, the reverse can also be true. For example, improvement in market efficiency has a significant effect on the market microstructure. In efficient markets, investors' perceptions of price are favorable, which generates positive signals for trading and stimulates liquidity. Failure to account for this bi-directional relationship can cause simultaneity bias (Arjoon, 2016). A possible way to cope with this issue is to include strictly exogenous variables in the model. This is not applicable to our case because most market microstructure variables are correlated with informational efficiency and, likewise, the error term. Therefore, instead of other models, we rely on generalized methods of moments (GMM). This method allows one to test the magnitude of a relationship in the absence of strictly exogenous variables. Arellano and Bond (1995) proposed using the first difference of all variables in the model to eliminate any potential bias due to unobserved heterogeneity. However, the lag estimator has its own shortcomings, e.g., exacerbated error terms and poor precision due to large sample finite bias. The modifications of Blundell and Bond (1998) are applied to cope with these issues, and therefore this study uses SGMM. This model integrates first-difference regressions with regression expressed in levels into a whole system. This method can mitigate bias in model specification due to endogeneity (Arjoon, 2016).

In regression analysis, we test the effect of microstructure on efficiency after controlling for both country-specific and stock-specific variables. The results are presented in five panels, showing the results for each market. Let us first highlight the effects of liquidity and efficiency. In our main analysis in Table 3, we assess liquidity with AMVST and STO. Our first major finding is that the beta coefficients for both measures of liquidity are negative and significant in all markets. However, the results are weakly significant in PSX and KLSE. The AMVST measure is better in terms of the frequency of significance in the analysis. It confirms that improvements in liquidity open doors to arbitrage activity and contribute to efficiency. This unmediated tie between efficiency and liquidity on Asian stock markets is consistent with Arias et al. (2022), Chordia et al. (2008) and Chung and Hrazdil (2010) and prove liquidity as a key factor in long-run dependence.

The analysis further shows negative beta loading, confirming the nature of the relationship between volatility and efficiency (only in PSX and KLSE). These results are in line with (Arjoon, 2016 and Ammar & Hellara, 2022). On the other hand, volatility (SSE, NSEI, and BSI) has positive and significant beta coefficients. Based on our results, excessive volatility contributes to inefficiency in the latter three markets which can be attributed to the fact that many economic aspects are now at their highest levels. Our results, showing excessive volatility causes inefficiency, are consistent with Lim and Kim (2011). Excessive volatility suggests that stock price changes are mainly noise-driven instead of providing accurate information about future earnings. A higher level of volatility is associated with greater return autocorrelations and hence a lower degree of efficiency. This

implies that market trading systems that aim to contain excessive volatility might deteriorate efficiency. However, the effect is more visible after controlling for idiosyncratic risk. It is interesting to note that the effect varies among all the sample countries. Volatility's effect on efficiency is more visible in China, followed by India and Turkey.

Thin trading, which proclaims that in this situation there are very few traders who deal in very small quantities for a very short period of time. This results in high fluctuation and causes noise trading that does not reflect true trade or the fundamental value of stocks and thus contributes to deteriorated efficiency. Thus, the few traders participating in buying and selling cannot represent the entire market and negatively affected efficiency. Overall, the link between anonymous trading and efficiency is significant with relatively high magnitude in SSE, NSEI, and BSI. The reason is that these markets are leading markets and handle trades in large quantities. This leads to high trading volume and increased liquidity, which promotes stock markets' informational efficiency.

The results of trading costs are consistent across markets (only weakly significant in NSEI and BSI). An inverse association shows that an increase in trading costs makes traders more likely to move away from fundamental values and join noise trading and reduce efficiency. The cost of trading decreases the informational value of stock prices, which exhibits an inverse relationship with market efficiency (Davila and Parlato, 2021). Since investors may perceive the value of this information to be lower than the level of trading costs, they are less interested in collecting it. This causes longer delays in stock price reactions to information, resulting in lower efficiency.

The weak positive coefficient suggests that automated trading slightly improves efficiency. This supports our hypothesis that electronic trading makes it easier for the market to incorporate new information into equity pricing. We attribute this finding to the increased trading speed due to the automated trading platform. Orders are matched and processed quickly, reducing the time it takes for information to be included in costs. The order-matching process makes it simpler for a broker to locate other traders to complete orders in the event once investor defaults on his transaction. As a result, trades are completed faster, which accelerates new information incorporation into prices. The application of information technology enables investors to access information. Automation is expected to reduce manual systems costs and inefficiencies, increase trading activity, and improve market transparency and liquidity on the stock markets. This is done by speeding up operations. The results show that tick size significantly affects efficiency (albeit the results are relatively weak in PSX and KLSE).

The results for financial liberalization in most cases are significant, albeit weak. A key reason is the embryonic stage of financial liberalization in these markets. The coefficient for investment freedom is relatively small, which indicates the lack of financial openness required to facilitate efficiency. Policy makers must take key steps to attract more foreign capital to liberalize these markets (Yang et al. 2022). Stock inherent risk also contributes to the market efficiency besides macroeconomic factors. Investors can mitigate idiosyncratic risks by diversifying their investment portfolios. Due to the extremely high level of idiosyncratic risk, the KLSE market is relatively less efficient. In this way, noise trading is possible, and it can agitate and limit arbitrage risks, which deteriorate efficiency.

We find that countries with higher uncertainty avoidance scores exhibit larger stock market declines than those with lower uncertainty avoidance scores. The IDRSK is another control variable with significant results. It is the inherent risk involved in investing in a specific asset, i.e., company securities. Idiosyncratic risk is particular to a specific investment and unlike systemic risk investors can mitigate it through diversification. Further, it is concluded that idiosyncratic volatility in stock returns predominantly reflects informational efficiency. When applying the SGMM model, it needs to apply two post-estimation tests to ensure a proper econometric model is employed. These tests are the Sargan test and the Arellano-Bond test for first-order and second-order correlation. Sargan and AR(1) and AR(2) tests for over-identifying restrictions are used. The Sargan test determines whether the econometric model is valid and the instruments are correctly specified. If the Sargan test turns out to be insignificant, this implies that the instruments included in the econometric specifications are exogenous. To examine the validity of a strong exogeneity assumption, the Arellano-Bond test of no auto-correlation is used under the null hypothesis that the error terms of two different time periods are uncorrelated.

5. Conclusion

This study investigates the relationship between market microstructure and informational efficiency in five Asian emerging markets namely China, India, Turkey, Pakistan, and Malaysia. We consider seven key components of market microstructure: liquidity, volatility, trading cost, anonymous trading, thin trading, automation, and tick size. The sample markets in this study differ in institutional, political, legal and macroeconomic conditions therefore we control for country level macroeconomic variables using investment freedom, uncertainty avoidance and GDP.

The empirical results, based on 2001–2020, revealed that market microstructure plays a vital role in improving emerging markets' efficiency. The multivariate regression confirms that the improvement in efficiency is robust even after controlling for country-specific variables (economic growth, investment freedom, and uncertainty avoidance) and stock-specific idiosyncratic risk. Study shows that liquidity is significant in all markets, while weakly significant (in PSX and KLSE) and Volatility (SSE, NSEI, BSI) has positive and significant beta coefficients. SSE, NSEI, and BSI show a significant link between thin trading and efficiency, while anonymous trading has a relatively low magnitude. The results of trading costs are consistent across markets (only weakly significant in NSEI and BSI). The weak positive coefficient suggests that automated trading slightly improves efficiency. The results show that tick size significantly affects efficiency (albeit the results are relatively weak in PSX and KLSE). The results for financial liberalization in most cases are significant, albeit weak. The coefficient for investment freedom is relatively small. Countries with higher uncertainty avoidance scores exhibit larger stock market declines than those with lower uncertainty avoidance scores. The IDRSK is another control variable with significant results.

This study is of particular importance to investors and policymakers. Investors need to account for the market microstructure and build their portfolios and investment strategies accordingly. The microstructure of the market must be closely monitored by individual and institutional investors. This has serious implications for stock selection and allocation decisions. Apart from these attributes, investors need to keep a close eye on the economic development, financial liberalization,

and culture of a country while implementing their portfolios. The findings also encourage investors to extend their investment horizons to international markets. To improve liquidity, policymakers should formulate mechanisms to further reduce transaction costs. The provision of timely, free of cost and reliable information, such as investor pamphlets and bulletins, can also reduce volatility and improve efficiency. Future studies are required to test the differential response of efficiency to microstructure components in an alternative sample of countries. Other avenues that ought to be explored are transparency, price discovery and manipulation, stealth trading, trading mechanisms and depository receipts, circuit breakers, short selling, earnings announcements, IPOs, and political conditions, as well as some financial reform variables such as different regimes of exchange and interest rates, variation in foreign trade quotas, changing country monetary and fiscal policies, macro announcements, and foreign portfolio flows.

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